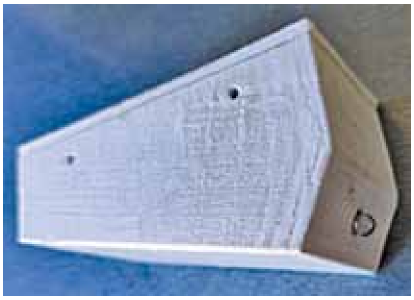

Fig. 12: Room ceiling mounted enclosure

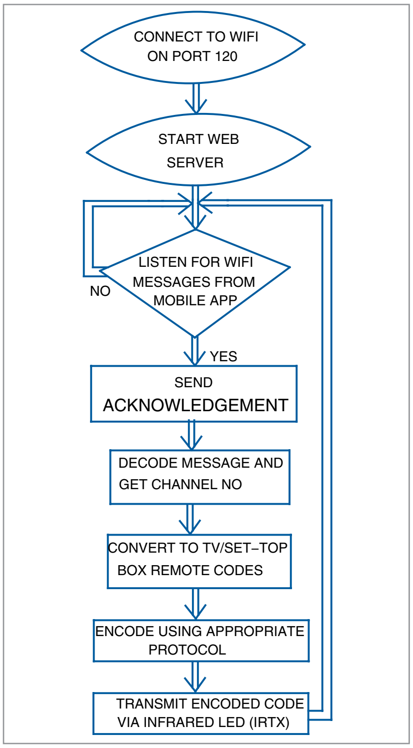

Fig. 13: Flowchart

The gateway address is the IP address of your router. Here, the first three values of the router address are the same as the first three bytes of the IP address of the NodeMCU.

The subnet address is the subnet mask defined for your local network. The DNS address is the domain name server address and can be seen in your router internet settings. Since each router has different router administration screens, you have to either Google the details or consult your network provider for help.

If you find the above two options difficult, you may remove the above lines from the code. The router will then allocate an IP address of its choice to the NodeMCU/IR-Blaster, which may vary each time the NodeMCU boots up. In such a case, you will have to set up the new IP address in the Settings screen of your TVRemote mobile app explained above.

PARTS LIST

Semiconductors:	
Board1, Board2	- NodeMCU development board
T1, T2	- 2N2222 NPN transistor
IR Receiver	- TSOP4838 IR receiver
IRTX1, IRTX2	- IR transmitter LED
LED1	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1, R2	- 10-ohm
R3-R5	- 1-kilo-ohm
Miscellaneous:	
Power supply	- 5V DC power supply

LED1 at pin D7 will light-up while the NodeMCU is connected to the network. After the Wi-Fi starts up, the NodeMCU will start the web server on port 120. The port number is an arbitrary number given to remove conflict with other programs using the standard http port 80. This completes the Setup( ) section of the sketch.

The loop( ) keeps listening to new incoming http (over Wi-Fi) messages from the mobile app. When it receives a message, it decodes it and extracts the channel number requested by the mobile app. To simplify the operation, only one type of message is sent.

Pseudo channel numbers have been assigned to operations like Power on/off, Volume plus/minus, etc. For a TataSky set-top box, channel numbers greater than 9000 have not been used; these free numbers have been assigned to such operations.

The incoming http message from the mobile is responded by the NodeMCU by sending a 200 response along with the channel number to the mobile app. The acknowledgement serves as a confirmation to the user that his request has been received by the IR-Blaster. If the mobile app fails to get this acknowledgement message within reasonable time, an error message showing 'IR Blaster not connected' is displayed on the mobile.

EFY Note
The source code of this project is available for free download at *source.efymag.com*

The sketch then checks whether the channel number received is within one of the following ranges:

- Between 0 and 9
- Between 100 and 8999
- Between 9001 and 9999

Single-digit channel numbers represent the digit keys that the user presses to type in any channel number. These keys are the equivalent of the number keys on your set-top box remote. IR code for each number pressed is transmitted in the sequence in which they are typed.

The second range (100 to 9000) represents the channel numbers assigned by TataSky to individual TV stations/channels. This feature is not available in the TataSky remote but has been built in this mobile Wi-Fi version. You can select any desired channel icon on the app and the set-top box will move to that channel. For example, on selecting DD News, the corresponding channel number 501 will be transmitted, and the user needs not press the 5, then 0, and then 1 key.

The third range (9001 to 9999) is used to signify special keys, like Volume plus, Volume minus buttons. The corresponding codes have been incorporated in the software (lines 103 to 140 of the sketch).

The IR message is encoded separately for the TataSky box (lines 222 to 220 of the sketch) and the TV (lines 229 to 232 of the sketch). Protocols of most of the common TV sets are built into the *IRremoteESP8266* library, which we also used for decoding explained earlier.

The author's prototype has been built for a Sony TV used in conjunction with a TataSky set-top box. Since readers may have a different brand of TV or set-top box, they need to modify the mobile app and the sketch to their particular brands using the author's code as a sample. Main changes in the app that may need to be done are the channel numbers assigned to different channels (as per the set-top box) and the IR codes and protocols that need to be changed in the sketch. EFY

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